

Overview of Image captioning

Subject: Image captioning

1.

Stages The process to generate text can be as simple as keeping a list of canned text that is copied and pasted, possibly linked with some glue text. The results may be satisfactory in simple domains such as horoscope machines or generators of personalized business letters. However, a sophisticated NLG system needs to include stages of planning and merging of information to enable the generation of text that looks natural and does not become repetitive. The typical stages of natural-language generation, as proposed by Dale and Reiter, are:

2.

Grass pollen levels for Friday have increased from the moderate to high levels of yesterday with values of around 6 to 7 across most parts of the country. However, in Northern areas, pollen levels will be moderate with values of 4.

3.

Automatic report generation From a commercial perspective, the most successful NLG applications have been data-to-text systems which generate textual summaries of databases and data sets; these systems usually perform data analysis as well as text generation. Research has shown that textual summaries can be more effective than graphs and other visuals for decision support, and that computer-generated texts can be superior (from the reader's perspective) to human-written texts. The first commercial data-to-text systems produced weather forecasts from weather data. The earliest such system to be deployed was FoG, which was used by Environment Canada to generate weather forecasts in French and English in the early 1990s. The success of FoG triggered other work, both research and commercial. Recent applications include the UK Met Office's text-enhanced forecast. Data-to-text systems have since been applied in a range of settings. Following the minor earthquake near Beverly Hills, California on March 17, 2014, The Los Angeles Times reported details about the time, location and strength of the quake within 3 minutes of the event. This report was automatically generated by a 'robo-journalist', which converted the incoming data into text via a preset template. Currently there is considerable commercial interest in using NLG to summarise financial and business data. Indeed, Gartner has said that NLG will become a standard feature of 90% of modern BI and analytics platforms. NLG is also being used commercially in automated journalism, chatbots, generating product descriptions for e-commerce sites, summarising medical records, and enhancing accessibility (for example by describing graphs and data sets to blind people). An example of an interactive use of NLG is the WYSIWYM framework. It stands for What you see is what you meant and allows users to see and manipulate the continuously rendered view (NLG output) of an underlying formal language document (NLG input), thereby editing the formal language without learning it. Looking ahead, the current progress in data-to-text generation paves the way for tailoring texts to specific audiences. For example, data from babies in neonatal care can be converted into text differently in a clinical setting, with different levels of technical detail and explanatory language, depending on intended recipient of the text (doctor, nurse, patient). The same idea can be applied in a sports setting, with different reports generated for fans of specific teams.

4.

Evaluation As in other scientific fields, NLG researchers need to test how well their systems, modules, and algorithms work. This is called evaluation. There are three basic techniques for evaluating NLG systems:

5.

Image captioning Over the past few years, there has been an increased interest in automatically generating captions for images, as part of a broader endeavor to investigate the interface between vision and language. A case of data-to-text generation, the algorithm of image captioning (or automatic image description) involves taking an image, analyzing its visual content, and generating a textual description (typically a sentence) that verbalizes the most prominent aspects of the image. An image captioning system involves two sub-tasks. In **Image Analysis**, features and attributes of an image are detected and labelled, before mapping these outputs to linguistic structures. Recent research utilizes deep learning approaches through features from a pre-trained convolutional neural network such as AlexNet, VGG or Caffe, where caption generators use an activation layer from the pre-trained network as their input features. **Text Generation**, the second task, is performed using a wide range of techniques. For example, in the Midge system, input images are represented as triples consisting of object/stuff detections, action/pose detections and spatial relations. These are subsequently mapped to <noun, verb, preposition> triples and realized using a tree substitution grammar. A common method in image captioning is to use a vision model (such as a ResNet) to encode an image into a vector, then use a language model (such as an RNN) to decode the vector into a caption. Despite advancements, challenges and opportunities remain in image capturing research. Notwithstanding the recent introduction of Flickr30K, MS COCO and other large datasets have enabled the training of more complex models such as neural networks, it has been argued that research in image captioning could benefit from larger and diversified datasets. Designing automatic measures that can mimic human judgments in evaluating the suitability of image descriptions is another need in the area. Other open challenges include visual question-answering (VQA), as well as the construction and evaluation multilingual repositories for image description.

6.

Content determination Deciding what information to mention in the text. For instance, in the pollen example above, deciding whether to explicitly mention that pollen level is 7 in the southeast. **Document structuring** Overall organisation of the information to convey. For example, deciding to describe the areas with high pollen levels first, instead of the areas with low pollen levels. **Aggregation** Merging of similar sentences to improve readability and naturalness. For instance, merging the sentences "Grass pollen levels for Friday have increased from the moderate to high levels of yesterday." and "Grass pollen levels will be around 6 to 7 across most parts of the country." into a single sentence of "Grass pollen levels for Friday have increased from the moderate to high levels of yesterday with values of around 6 to 7 across most parts of the country." **Lexical choice** Putting words to the concepts. For example, deciding whether medium or moderate should be used when describing a pollen level of 4. **Referring expression generation** Creating referring expressions that identify objects and regions. For example, deciding to use in the Northern Isles and far northeast of mainland Scotland to refer to a certain region in Scotland. This task also includes making decisions about pronouns and other types of anaphora. **Realization** Creating the actual text, which should be correct according to the rules of syntax, morphology, and orthography. For example, using will be for the future tense of to be.

An alternative approach to NLG is to use "end-to-end" machine learning to build a system, without having separate stages as above. In other words, we build an NLG system by training a machine learning algorithm (often an LSTM) on a large data set of input data and corresponding (human-written) output texts. The end-to-end approach has perhaps been most successful in image captioning, that is automatically generating a textual caption for an image.